

A Portable and Low-Cost Single Board Computer-Based Spectrophotometric Platform for Optical Analysis in the UV and Visible Range

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The growing ability of high-performance consumer electronics is fostering innovation in prototyping and research. Although the Raspberry Pi single-board computer (SBC) is widely used across various fields, including diagnostics, applications involving UV-based reading remain underexplored. In this study, a simple, low-cost (less than 100 euros), and portable device for UV–vis measurements based on Raspberry Pi is presented. The system integrates the AS7341 multi-channel multispectral digital sensor, offering analytical performance comparable to two commercial spectrophotometers – one benchtop and one portable – in terms of linearity, limit of detection, quantitation, and repeatability. Furthermore, the device demonstrates reliable performance both in analyzing real wine samples using two different commercial enzymatic kits and in quantifying different concentrations of synthetic DNA oligonucleotides. By providing a ready-to-use, portable, and affordable platform for UV–vis diagnostics, this prototype is well-suited for resource-limited settings, educational environments, and IoT applications where numerous distributed sensors are required for specific diagnostic tasks.

microcontrollers, such as the Arduino or Adafruit board ecosystems, and SBCs such as Raspberry Pi, has inspired a wide range of applications, spanning research, device customization, prototyping, and education. Over the past decade, Raspberry Pi has made a significant impact across numerous disciplines. A search of major scientific databases reveals over 400 articles focused on Raspberry Pi, presenting applications spanning a wide range of solution types, including education, robotics, healthcare, agriculture, energy management, image processing, astronomy, telecommunication, automation, surveillance, industrial, nanotechnology, and diagnostics.^[6] Raspberry Pi, developed by the Raspberry Pi Foundation, is a low-cost, credit-card-sized computer aimed at promoting computer science education and fostering STEAM learning

1. Introduction

Microcontroller boards (MCBs), single-board computers (SBCs), and other consumer electronics have become essential tools in research, especially in the development of imaging, sensing, and diagnostics technologies.^[1–5] The fast-growing accessibility of

in both children and adults.^[7] Various models of this SBC, including Zero, ZeroW, ZeroWH, A/A+, 2, 3, 3B/3B+, 4, 4B, and the recent Raspberry Pi 5, cater to different user needs and budgets. Several distinctive features of Raspberry Pi make it ideal for diagnostics and laboratory applications: i) comprehensive input/output signal routing; ii) compatibility with numerous external modules; iii) support for standard communication protocols (e.g., USB, Ethernet, Wi-Fi, Bluetooth, TX/RX, I²C, SPI) enabling seamless integration with external devices; iv) a broad range of available programming languages (C, C++, Java, Javascript, Ruby, Python, R, C#) and integrated development environments (IDEs), facilitating programming, porting and learning curve for beginners; v) a large, active community of users offering open-source projects and troubleshooting resources, making self-learning fast and accessible. In recent years Raspberry Pi boards have been deployed in a wide array of diagnostic applications, including real-time polymerase chain reaction (PCR), digital PCR and isothermal amplification methods,^[8–11] microfluidic systems for pathogen detection and cell growth monitoring,^[12] surface plasmon resonance (SPR) spectrometers,^[13] multi-well plate readers,^[14] portable photometers and spectrophotometers,^[15,16] lab-on-a-chip systems for reduced sample volume handling,^[17] atomic force microscopes,^[18] optical and fluorescence microscopy,^[19,20] gas

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monitoring automation,^[21] laboratory automation^[22,23] and plant phenotyping.^[24] However, a review of the literature reveals that diagnostic devices based on Raspberry Pi have so far been limited to detecting visible light or infrared targets. These studies typically utilize sensors equipped with RGB filters and lenses to capture conventional digital images for signal processing and analyte quantitation. To date, the only reported use of Raspberry Pi-based devices with UV radiation focused on sulfur dioxide detection in industrial and volcanic sources, however without quantitative measurements.^[25] Given the wide availability of open-source software, hardware, and high-performance optoelectronic components for Raspberry Pi, this gap in UV diagnostics is a significant limitation. UV-based detection is particularly valuable for applications involving enzymatic kits typically used in clinical, food, and industrial diagnostics. In response to this, we explored the AS734, a multi-channel colorimetric sensor compatible with Raspberry Pi, and evaluated its potential as a single detector for both visible and UV light-based applications. We developed a simple, low-cost, miniaturized diagnostic device by integrating AS7341 with Raspberry Pi, and demonstrated that its analytical performance is comparable to two commercial spectrophotometers. The prototype was tested on real wine samples using two commercial enzymatic kits and for quantifying various concentrations of synthetic labeled DNA, providing reliable results that showcase its potential for enabling UV-vis diagnostic measurements across diverse settings.

2. Results and Discussion

2.1. Device Settings and Optimization

One of the key advantages of the Raspberry Pi platform is its seamless connectivity with a wide variety of optical sensors, providing out-of-the-box solutions that include circuitry, cabling, and dedicated libraries for scripting in languages such as Python. Previous work extensively investigated the use of the Raspberry Pi camera (PiCam, v1.3 board) mounting the Omnivision OV5647 sensor as a detector for absorbance measurements.^[27,28] After reviewing several low-cost consumer optical sensors mounted on the Breakout Board (including GUVA-S12SD, LTR390, SI1145, and the 6-Channel AS7262), we concluded that the AS7341 was the optimal choice for developing a prototype device designed to: i) maximize detectable wavelengths; ii) combine UV and visible light reading capability; iii) provide high-end electronic performances. The AS7341 is an 11-channel spectrometer suited for consumer, commercial, and laboratory applications such as spectral identification, reflection and absorption analysis for color matching, fluid or reagent analysis, passive ambient light measurement, and color calibration.

The AS7341 spectral response range is defined by individual channels covering ≈ 350 –1000 nm region, with eight channels in the visible spectrum (VIS), centered on 415, 445, 480, 515, 555, 590, 630, and 680 nm, along with one channel for near-infrared (NIR) and a clear channel for simultaneous detection capabilities. Light source detection can also be supported by an integrated additional flicker channel. The sensor is housed in an ultra-low-profile package measuring 3.1 mm $\times$ 2 mm $\times$ 1 mm. The device is controlled and monitored via registers accessed through the

I²C serial interface, allowing for both control functions and reading data from the sensor. The device features a 4 $\times$ 4-photodiode array geometry. Above and below this array are two additional photodiodes with dedicated functions such as flicker detection ("FLICKER") and near-infrared response ("NIR"). The sensor also includes a clear channel ("C") photodiode without a filter in the left and right bottom corners (Figure 1).

The signals from each photodiode are routed through a 16-bit 6-channel Analog-to-Digital (ADC), which converts raw measurements into digital values that can be read via I²C, providing adequate performance for diagnostic purposes. While the AS7341 has been proposed for use as a visible light detector in a recent study,^[29] it has never been proposed for UV analysis exploiting the "clear channel" as a general-purpose detector. For the subsequent experiments, 3D-printed holders were designed to fit the sensor and the various LEDs with different form factors (Figure 2).

The dimensions and positioning of the components were kept consistent across all setups, and no additional optical components were introduced. Although the Raspberry Pi lacks a digital-to-analog converter (DAC) interface to control the LED intensity, for LEDs in the visible region, the intensity was regulated through pulse-width modulation (PWM) digital control via GPIO18, allowing control of the duty cycle between 0 and 1024 values. The response of the sensor to varying LED signals was tested by adjusting LED intensities (Figure 3).

For all the tested channels, the LED-detector combination demonstrated good linearity and signal intensity, making calibration of the prototype's light source straightforward and reliable for analytical purposes. The stability of the system was assessed by reading the source signal every 20 min over a 4 h period, with the LED intensity set to $\approx 20\%$ of the 16-bit full scale (65535). The platform showed comparable performance to commercial devices, with a coefficient of variation (CV%) of less than 1.00. While we note that also an in-depth characterization of the NIR channel would be interesting, this falls outside the scope of our UV-vis-focused study. The NIR region near 900 nm has limited relevance for ready-to-use kits, the primary application area of this work. Additionally, the sensor's inaccuracy in resolving individual wavelengths within the broad NIR range limits its use in model-based NIR spectroscopy.

2.2. Device Characterization

The ability of the prototype to measure the absolute absorbance of target molecules was evaluated both in the visible and the UV regions of the spectrum. For the visible region, measurements were conducted using several dilutions of a red food dye, with signals acquired at $\lambda = 515$ nm. We selected a dye similar to these used in previous works aiming to evaluate the PiCam as a photometric sensor. The results demonstrated an excellent correlation between the prototype and the conventional benchtop spectrophotometer used as a reference instrument (Figure 4A).

It is noteworthy that the linearity of the prototype extends to absorbance values significantly higher than 1.00, a value traditionally considered a critical upper limit for linear response. While this is not universally applicable, it is particularly valuable in modern diagnostics, where instruments capable of maintaining

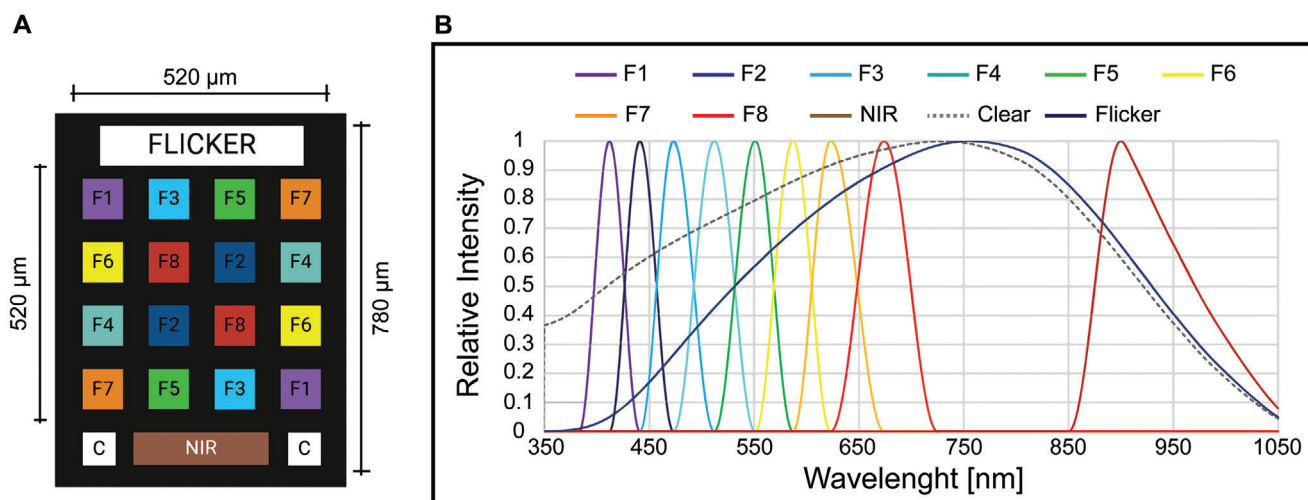


Figure 1. A) Spatial arrangement of the 4×4 -photodiode array of the AS7341 sensor; B) Normalized spectra responsivity for each channel. (Source: Product Document ams OSRAM Group, DS000504 • v3-00 • 2020-Jun-25).

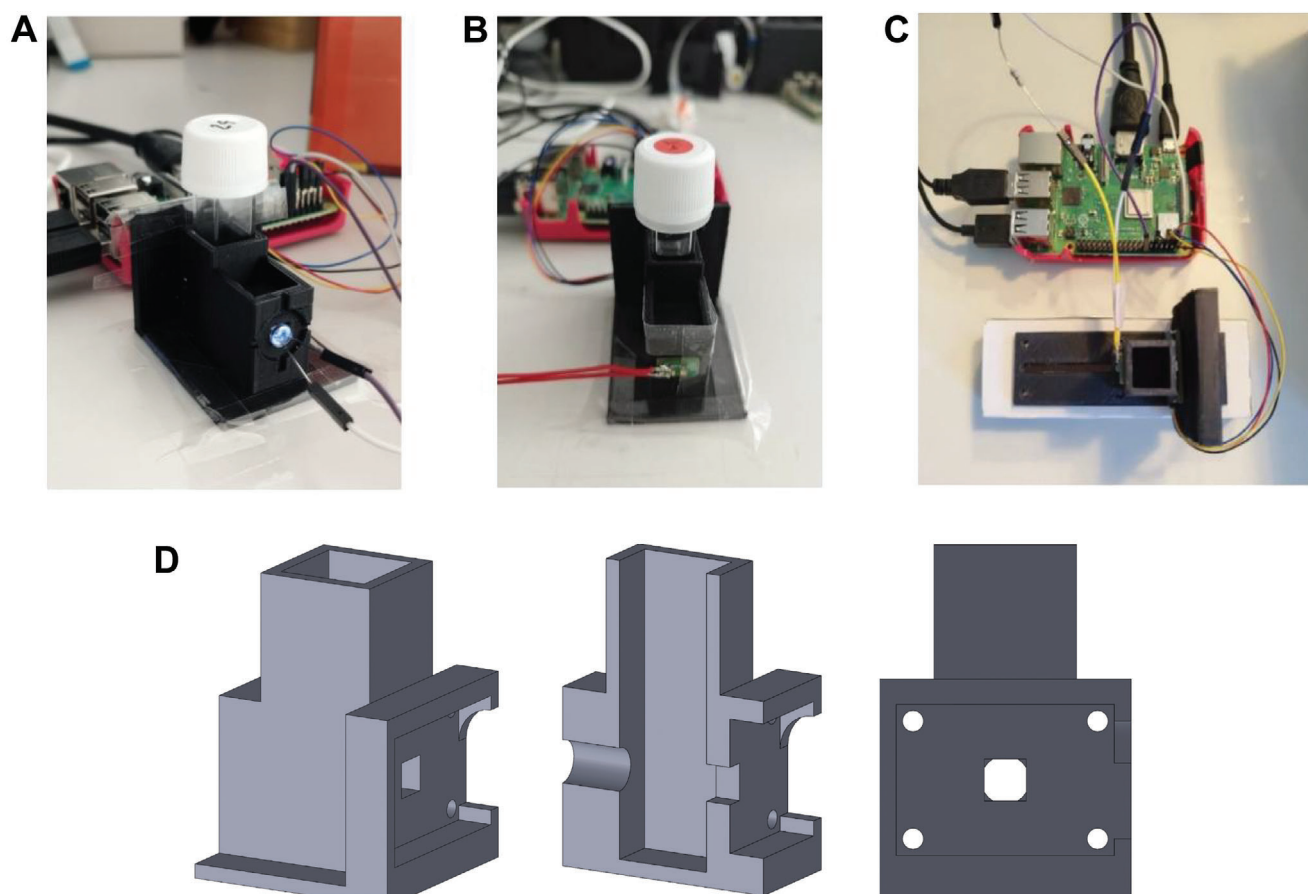


Figure 2. A, B) 3D-printed setups used for testing with visible LEDs; C) Setup designed for the UV experiments. All setups share consistent characteristics, differing only in their LED-fitting shape; D) Schematic representation of the 3D-printed components of the detection system, shown from multiple perspectives.

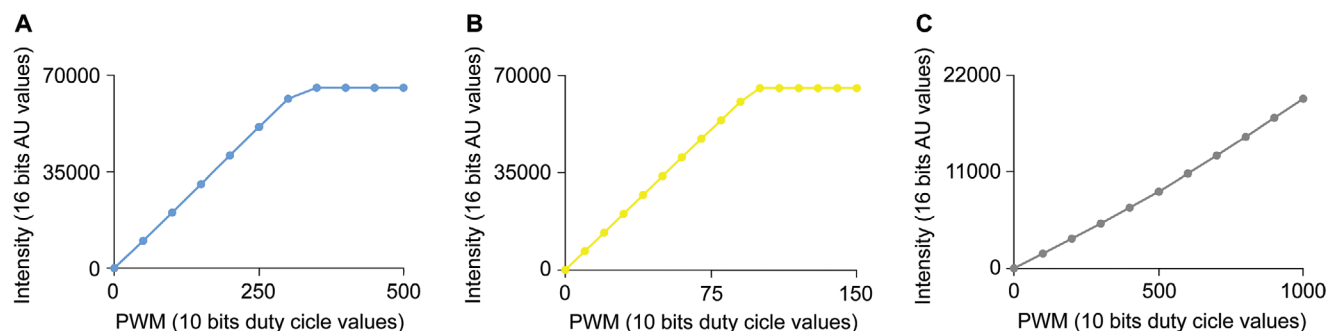


Figure 3. Signal intensity of the tested sensor channels as a function of PWM values. A) F4 channel (515 nm); B) F6 channel (590 nm); C) Clear channel (no filter). Employed LEDs were centered on 505, 590, and 340 nm respectively. Channel names are reported according to the AS7341 sensor product document.

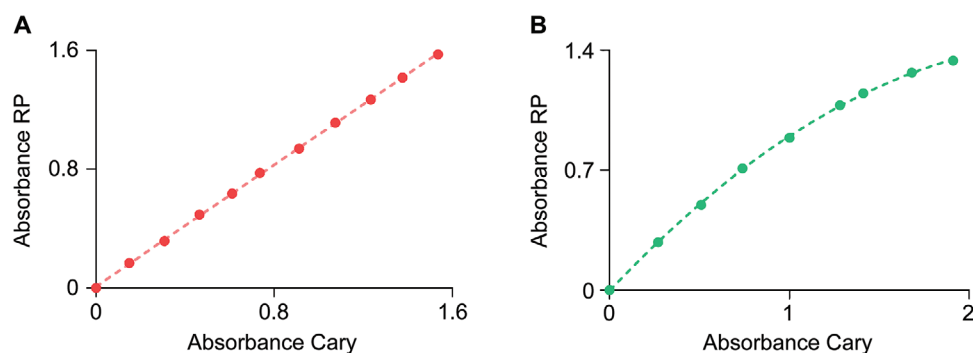


Figure 4. Correlation between absorbance calculated using the Raspberry Pi prototype and the benchtop spectrophotometer. A) Red food dye acquired at $\lambda = 515$ nm. Equation: $y = 1.024x + 0.009546$ ($R^2 = 0.9998$); B) Dilution of NADH solution acquired at $\lambda = 340$ nm. Equation: $y = -0.2150x^2 + 1.118x - 0.004849$ ($R^2 = 0.9998$).

a linear response at high absorbance values are highly desirable. This feature allows for the use of ready-to-use kits without the need for dilution steps, simplifying the analytical process within the desired range. In the UV region, dilutions of NADH were employed to assess the performance of the sensor using the clear channel (Figure 4B). NADH was chosen for this evaluation as its peak absorbance at 340 nm falls within the region of interest, it is a well-characterized molecule, stable, easy to handle, and highly relevant for commercially available enzymatic test kits that utilize NADH or NADPH. Interestingly, the Raspberry Pi prototype showed an excellent ability to discriminate absorbance signals above 2.30 but began to deviate from linearity around an absorbance of 1.00. This deviation in the prototype's absorbance calculation could be attributed to two factors: i) an initial I_0 (reference light source intensity) reading higher than expected, causing the sensor to detect lighter than is actually present, leading to an artefact starting signal, or ii) a smaller I (sample intensity), where less light reaches the sensor after the sample is introduced. The latter seems more relevant, as it corresponds with the observation that this phenomenon becomes more pronounced at higher absorbance levels, when light intensity is extremely low (10% to less than 1% of the initial value). The optical cause behind this behavior may be related to the uncomplicated design of the prototype which intentionally avoids the use of additional components such as lenses, grating, and other common elements available in traditional spectrometers. The absence of elements that

collimate and focus the light source, combined with limitations in the 3D printing process, may have resulted in unintended light dispersion. Additionally, it should be noted that the responsivity of the photodiode associated with the clear channel in the presence of UV signals is not detailed in the product technical documentation of the AS7341 sensor. Despite this deviation, the sensor proved entirely suitable for the original purpose of the study. All subsequent tests were carried out after linearizing the clear channel response, using correction factors derived from the correlation in Figure 4B.

2.3. Platform Validation

To evaluate the diagnostic capabilities of the device in the UV region, preliminary validation tests were performed. The first stage of validation involved using a commercial ready-to-use enzymatic kit combined with its dedicated reader, following the manufacturer's instructions. The same reagents were then applied to the same samples using a general-purpose kit with a third-party spectrophotometer. We selected the enzymatic test for the total amount of D-Glucose/D-Fructose in food matrices for several reasons: i) it is relevant across multiple fields; ii) it is robust and easy to perform, minimizing operator-related variability; and iii) it features a high linearity range and requires high-performance photometers capable of absorbance readings up to 2.50 absorbance units. The first test involved verifying

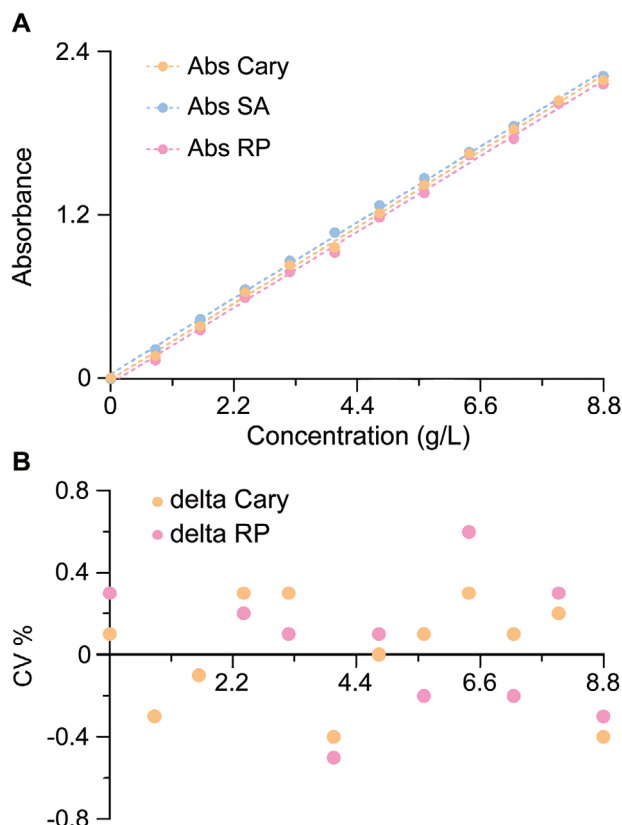


Figure 5. Comparison of the calibration curve for the D-Glucose / D-Fructose enzymatic kit using the Raspberry Pi prototype, a general-purpose benchtop spectrophotometer (Cary 60), and a dedicated instrument (Smart Analysis). A) Regression analysis Equation Cary: $y = 0.2547x - 0.01244$ ($R^2 = 0.9986$), Equation Smart Analysis: $y = 0.2536x + 0.02756$ ($R^2 = 0.9989$), Equation Raspberry Pi prototype: $y = 0.2535x - 0.04205$ ($R^2 = 0.9980$); B) Residual analysis.

the linearity range declared by the commercial kit using the Raspberry Pi prototype (Figure 5). For this purpose, 12 samples (including a blank) of a D-Glucose / D-Fructose standard solution were used to generate a calibration curve. Each point was measured at the end of the reaction using the three instruments. The prototype demonstrated the ability to support the nominal range of the kit up to 8.0 g L^{-1} , with regression coefficients showing an excellent correlation across all three instruments.

The repeatability of the system was assessed by reading ten replicates (concentration $\approx 4.00 \text{ g L}^{-1}$) during a single analytical session over 60 min. The mean concentrations measured were 3.87 g L^{-1} for the prototype and 3.85 g L^{-1} using the Agilent Cary 60, indicating a comparable bias. Notably, the standard deviation was 0.07 g L^{-1} for both platforms, highlighting the strong repeatability of the prototype within the timeframe of a typical analytical session. Finally, the limit of detection (LOD) and limit of quantification (LOQ) were estimated using the calibration curve method (Table 1), with three concentrations tested over six replicates for each level. Both the LOD and the LOQ of the prototype were comparable to the values obtained from the two commercial instruments.

Table 1. Estimation of the limit of detection (LOD) and limit of quantification (LOQ) for the three devices: Smart Analysis (SA), Cary 60 (Cary), and Raspberry Pi prototype (RP).

Device	LOD [g L^{-1}]	LOQ [g L^{-1}]
SA	0.10	0.35
Cary	0.07	0.21
RP	0.1	0.30

2.4. UV Analysis in a Complex Food Matrix

The preliminary characterization of the prototype for potential UV applications indicated performance comparable to the two commercial instruments. To validate this in a real-world laboratory context, a comparative analysis using real samples was conducted. A dataset of 44 wines (red, white, and rosé) was analyzed for the quantitative determination of D-Glucose / D-Fructose content using the ready-to-use kit employed in the validation phase and another manual kit from R-Biopharm, this latter regarded as a reference product in the field. All samples were measured using the Raspberry Pi prototype alongside the two reference instruments. The comparative results are presented in Table S1 (Supporting Information). Overall, the Raspberry Pi prototype delivered satisfactory results, closely aligning with the reference instruments. When examining the performance within each enzymatic method, a slight overestimation was observed with the prototype when paired with the DNAPhone kit. This discrepancy likely stems from differences in calibration accuracy. The DNAPhone kit is calibrated on commercial instruments using a multi-point calibration with five replicates and certified reference material, whereas the prototype employed a single calibration run with an aqueous standard. The absence of this overestimation with the R-Biopharm kit supports this hypothesis. For the R-Biopharm kit, concentrations were calculated directly from absorbance readings using the NADH molar extinction coefficient, indicating that the prototype is highly accurate in reading absorbance from enzymatic reactions. Additionally, we note that higher values were obtained for the R-Biopharm kit, particularly for highly concentrated samples, regardless of the instrument used. This trend appeared to be independent of the measuring device and is likely due to differences in the dilution approach. To obtain measurable results, the R-Biopharm kit required higher dilution of the samples, and as this step was critical for the final result, slight variations in the dilution could have impacted the outcome. To select the most informative comparative test, we evaluated the two methods that are conceptually the most divergent: the Raspberry Pi prototype combined with the DNAPhone kit, and the R-Biopharm paired with the Agilent Cary 60 spectrophotometer (the two gold standards among our options). For this purpose, non-parametric Passing-Bablok regression analysis and Bland-Altman analysis were performed (Figure 6).

The results from the regression analysis, including the upper and lower confidence intervals for the regression coefficients, showed no significant differences between the two methods, confirming that the Raspberry Pi prototype is suitable for analytical screening purposes. Additionally, the Bland-Altman analysis revealed a small bias with no apparent effects due to differences in

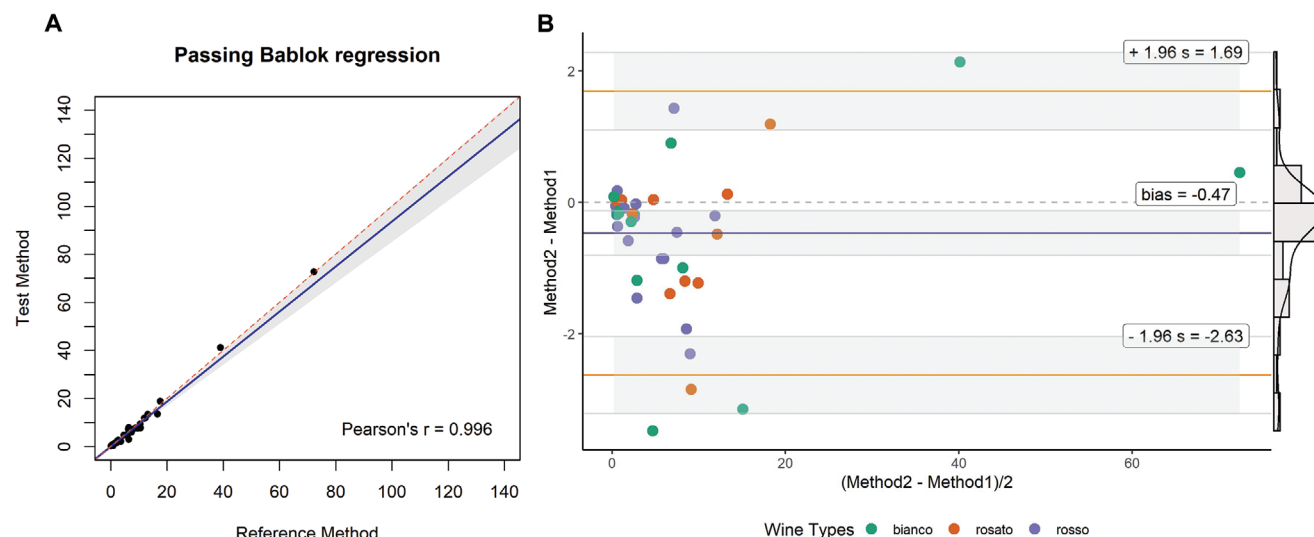


Figure 6. Comparative analysis for the two selected methods (DNAPhone kit with Raspberry Pi prototype vs R-Biopharm kit with the Cary 60 spectrophotometer). A) Passing-Bablok regression analysis for 43 wine samples. Intercept: -0.0568 (LCI: -0.1941 ; UCI: 0.0991); Slope: 0.937 (LCI: 0.853 ; UCI: 1.007). B) Bland-Altman plot.

the matrix (e.g., wine color or type), further demonstrating the robustness of the platform (Figure 6B).

2.5. Analysis of Synthetic DNA in the Visible Range

Given the successful results in both the visible and UV ranges, we conducted an additional experiment using fluorescently labeled synthetic DNA to explore further applications for the prototype. Quantifying synthetic DNA oligonucleotides by absorbance measurement is a routine task in research laboratories across various life science fields. For this experiment, fluorescein-labeled synthetic oligonucleotides, prepared at concentrations ranging from 1.0 to $10.0 \mu\text{M}$, were measured using both the Cary 60 spectrophotometer and the Raspberry Pi prototype. This concentration range was selected to achieve two main objectives: first, to produce measurable absorbances up to 1.0 ; and second, to assess any potential interference from fluorescence emissions due to the detector's proximity to the cuvette. As shown in Figure S1 (Supporting Information), the prototype demonstrated an excellent correlation ($R^2 = 0.9998$) with the spectrophotometer, with no observed deviation from linearity. These results suggest promising potential for the prototype in quantifying biologically relevant molecules, such as fluorophore-labeled DNA utilized in numerous diagnostic applications.

3. Conclusion

In conclusion, we successfully developed a simple, portable, and cost-effective diagnostic device based on Raspberry Pi, with a total construction cost of less than 100 euros. By integrating a multi-channel, multi-spectral digital sensor with Raspberry Pi's native connectivity, we created a versatile platform for photometric applications operating across both the visible and UV regions. Notably, to the best of our knowledge, no previous investigations

have utilized similar setups for UV applications, making this a novel approach in the field. The suitability of the device for real enzymatic applications was thoroughly explored and validated. Our results demonstrate that the Raspberry Pi prototype performs comparably to two commercial systems—a benchtop spectrophotometer and a portable device—in terms of analytical parameters such as linearity, limit of detection, limit of quantitation, and repeatability. The device exhibited excellent agreement with reference instruments when applied to two different commercial enzymatic kits for the quantitative analysis of D-Glucose/D-Fructose in real wine samples. These findings highlight the potential of this platform for accurate and reliable diagnostic measurements. The key advantages of this prototype include its portability, ease of use, and affordability. The absence of additional optical components, such as lenses or gratings, greatly simplifies the design without sacrificing performance, while the availability of open-source software libraries ensures broad accessibility for development and customization. Moreover, the Raspberry Pi's native connectivity features make it an attractive option for applications in resource-limited settings, educational environments, and IoT-based systems requiring a large network of distributed sensors. The presented device offers a promising solution for low-cost, high-performance diagnostic tasks and could serve as a practical tool in various fields, from academic research to on-site diagnostic screening, particularly in scenarios in which traditional, expensive instrumentation may not be feasible.

4. Experimental Section

Raspberry Pi Settings and Programming: The SBC Raspberry Pi 3 Model B+ was employed as the computing unit throughout all experimental sessions. The AS7341 sensor (ams-OSRAM AG, Austria) mounted on a commercial Breakout Board (Adafruit Industries, LLC, US) was plugged into the SBC through a QT / Qwiic JST SH 4-pin cable and the standard 4-Pin I²C cable routing was followed: Pi 3V to sensor VCC (red wire); Pi GND to sensor GND (black wire); Pi SCL to sensor SCL (yellow wire);

Pi SDA to sensor SDA (blue wire). The experiments were conducted using a standard Linux-based operating system Pi OS (ver 11, kernel 6.1.21), with Windows laptops connecting to the Raspberry Pi-based device via the standard SSH protocol. The scripts employed during the experiments were developed using Python (ver. 3.9.x) and the sensor was controlled using the Adafruit CircuitPython AS7341 library, following the basic instructions available on the GitHub page.^[26] The control library allows the adjustment of several parameters, including integration time, gain, channel selection, and the status of the on-board LED. The Python scripts used during the experiments were adapted from the manufacturer's example codes and executed directly from the Linux terminal. Data were saved as csv data and further elaborated separately using spreadsheet editors or programming languages (Python, R). Examples of the codes used in the analysis are reported in the [Supporting Information](#) file. For the visible light wiring setup, the LED was connected to GPIO 12 (power) and GPIO 14 (ground); through these connections it was possible to control the intensity of the LED through the PWM regulation. For the UV tests, the LED was connected to the GPIO 2 (5V power) and GPIO 14 (ground). A resistor was added to the circuit to accommodate the higher voltage requirements of the selected LED.

Device Assembling: Various simple 3D-printed setups were used for the measurements. Each configuration consisted of a holder designed to ensure the correct positioning and alignment of the AS7341 sensor, the selected LED, and a light diffuser. The two main setups were similar in terms of component distances, arrangement, and geometry, and were designed to properly house LEDs of different shapes and sizes (5mm round LEDs, SMD-mounted LEDSS, and other mini-LEDs). All 3D prototypes were designed and fabricated in-house by means of a 3D printer Sharebot Next Generation (Sharebot, Nibionno, LC, Italy) with polylactic acid (PLA) polymers.

Sample Preparation and Treatment: Samples for preliminary characterizations were prepared through dilutions of food dyes derived from natural plant-based ingredients (Rebecchi Food Systems S.p.A., Italy) in double-distilled water. Stock solutions were used after centrifugation at 5500 rpm for 5 min and stored at 4 °C for a week. Absorbance was measured by means of a Cary 60 spectrophotometer (Agilent Technologies, Santa Clara, CA, USA). The dye stock solutions were prepared to have an absorbance between 0.1 and 2.0. For the first experimental tests in the ultraviolet range and for the validation of the method, Titrivin standards, a well-known reference material for oenology laboratories, were used. The specific reference employed in this study was AA2 (lot number is A 04181222 2). For the comparative analysis, 44 commercial wines from different Italian wineries were purchased locally. The wines were degassed by sonication, centrifuged at 5500 rpm for 10 min and stored at 4 °C. All wine samples were used within one month of collection.

Enzymatic Reagent Kits for D-Glucose / D-Fructose Quantitative Determination: Two different ready-to-use kits from different manufacturers were employed for testing. First, the analyses using the kit (Ref RPF-REVGF-0020) from DNAPhone s.r.l (Parma, Italy) were carried out according to manufacturer's instructions provided in the instrument android app. Briefly, 10 µL of wine was added to a pre-filled, screw-capped cuvette containing 1500 µL of a proper buffer (R1). After incubation at 37 °C for 3 min, the first absorbance value (A1) was recorded at $\lambda = 340$ nm. Next, the reagent (R2) containing enzymes and coenzymes (Hexokinase, Glucose-6-phosphate isomerase, Glucose-6-phosphate dehydrogenase, NADPH, ATP) was added, followed by incubation at 37 °C for 7 min. After the incubation, the reaction endpoint was evaluated by reading a second absorbance value (A2). Second, the analyses using the kit (REF. E8160) from R-Biopharm (Pfungstadt, Germany) were carried out according to the manufacturer's instructions included in the package. Briefly, 2000 µL of "reagent 1" and 100 µL of samples were added to plastic (PMMA) cuvettes and incubated at 37 °C for 1 min. The first absorbance value (A1) was recorded for each sample, and the additional reagents ("reagent 2" and "reagent 3") were added. After incubation at 37 °C for 7 min, a second absorbance value (A2) was recorded. Both kits were tested by using the Cary 60 spectrophotometer (Agilent Technologies, Santa Clara, CA, USA), the DNAPhone proprietary platform Smart Analysis, and the Raspberry Pi-based prototype developed in this work.

Absorbance and Analyte Concentration Calculations: The Cary 60 spectrophotometer was used as a reference instrument for measuring sample absorbance. The manufacturer's standard software for UV-vis applications in scan mode was used throughout the experiments. For the Raspberry Pi prototype, raw intensities from the ten channels of the detector were selected and used for absorbance calculations. The absorbance (A) was calculated from the channel of interest using the equation:

$$A = \log_{10} \frac{(I_0 - I_d)}{(I_s - I_d)} \quad (1)$$

where I_0 is the input light intensity of the LED source, I_d is the residual signal after the LED is turned off and I_s is the intensity measured after the sample is introduced.

The concentration of analytes in the wine samples was calculated as following:

- 1) DNAPhone device/kit: the analyte concentration (g L^{-1}) was provided directly at the end of the analysis by the graphical Android app.
- 2) R-Biopharm kit: the concentration was calculated according to the manufacturer's instruction using the equation:

$$\Delta A = (A_2 - df * A_1)_{\text{sample}} - (A_2 - df * A_1)_{\text{blankBR}} \quad (2)$$

where df is the dilution factor, calculated as:

$$df = \frac{\text{sample volume} + R1}{\text{test volume}} \quad (3)$$

where R1 is the volume of reagent 1 (NAD).
and,

$$C_{\text{total D-Glucose/D-Fructose}} [\text{g/L}] = \frac{V * MW * \Delta A}{\epsilon * d * \nu * 1000} \quad (4)$$

where:

V: Test volume basic application [mL] = 3.100

MW: Molecular weight [g/mol] = 180.16

d: Optical path [cm] = 1.00

ϵ : Extinction coefficient NADH [L/mmol x cm] = 6.3 (at 340 nm)

- 1) Raspberry Pi prototype: Sample concentration was calculated by creating a multi-point calibration curve using aqueous standards.

Statistics: The data and the images were analyzed using Prism8 (GraphPad). The Passing-Bablok regression and the Bland-Altman analysis were performed using R version 4.3.1. In particular the mcr (ver 1.3.3) and the BlandAltmanLeh (ver 0.3.1) libraries were employed. The script was edited using RStudio (ver. 2024.09.0 + 375).

Supporting Information

Supporting Information is available from the Wiley Online Library or from the author.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Keywords

consumer electronics, diagnostic devices, low-cost devices, optical sensors, raspberry pi, UV–vis spectroscopy

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